Student Performance Prediction

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Abstract*—Predicting a student's academic performance in advance is a key enabler of early intervention and personalized educational support. This paper presents a complete end-to-end machine learning pipeline for predicting a student's writing performance category (Low, Good, or Excellent) from demographic, socioeconomic, and academic attributes, without using the writing score itself as an input, so as to avoid target leakage. The publicly available Students Performance dataset (1,000 records, 8 attributes) was analyzed through exploratory data analysis, feature engineering, and the training of five supervised classification algorithms: Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors (KNN), and Support Vector Machine (SVM). Each model was evaluated on a held-out 20% test set (200 students) using accuracy, precision, recall, and macro F1-score, and Logistic Regression was further validated using 5-fold stratified cross-validation and tuned using GridSearchCV. Logistic Regression achieved the best overall test performance, with an accuracy of 89.0% and a macro F1-score of 0.8892, outperforming Random Forest (88.0%), KNN (87.0%), SVM (86.0%), and Decision Tree (83.0%). Coefficient analysis of the selected Logistic Regression model showed that gender, test preparation course completion, and parental level of education were the most influential predictors of writing performance. The final model was applied to an unseen student record, predicting an Excellent writing performance category with 82.53% confidence. The results demonstrate that simple, interpretable linear models can be highly effective for structured educational data of this nature.

***Index Terms*—student performance prediction, educational data mining, machine learning, classification, logistic regression, feature importance, exploratory data analysis, model comparison**

I. Introduction

Educational institutions increasingly rely on data-driven methods to understand and improve student outcomes. The ability to anticipate a student's academic performance before final results are produced allows teachers and administrators to identify at-risk students and allocate support resources more effectively. Machine learning offers a systematic way of uncovering the relationships between a student's background, environment, and academic performance, moving beyond simple descriptive statistics toward predictive, actionable insight.

This paper documents a complete end-to-end machine learning project built on the well-known *Students Performance* dataset, which records the mathematics, reading, and writing scores of 1,000 students together with several demographic and socioeconomic attributes. Rather than predicting a continuous score, the project frames the task as a multi-class classification problem: predicting whether a student's writing performance falls into a *Low*, *Good*, or *Excellent* category. This framing produces a decision-support output that is easier for educators to act upon than a raw numeric score.

The remainder of this paper is organized as follows. Section II states the problem addressed. Section III and IV present the research objectives and research questions. Section V reviews related work, and Section VI identifies the research gap this study addresses. Section VII describes the dataset. Sections VIII–X describe the overall methodology, workflow, and preprocessing. Section XI presents the exploratory data analysis. Section XII discusses feature engineering and selection. Sections XIII–XVIII cover model development, experimental setup, validation, hyperparameter tuning, evaluation, and comparison. Section XIX describes the final model selection, and Section XX demonstrates prediction on unseen data. Sections XXI–XXIII present the results, discussion, and limitations. Sections XXIV–XXV present the conclusion and future scope, followed by references.

II. Problem Statement

Traditional academic reporting is retrospective: performance categories are only known after assessments are completed and graded. Educators and institutions, however, benefit most from performance indicators that can be estimated *before* or *independently of* a specific score, using information that is already available, such as demographic background, parental education, lunch program participation, test-preparation status, and performance in related subjects. The problem addressed in this paper is therefore: *given a student's demographic, socioeconomic, and academic attributes (excluding the writing score itself), can a supervised machine learning model reliably classify the student's writing performance into Low, Good, or Excellent categories?* A secondary problem is identifying which of several common classification algorithms is best suited to this structured, mixed-type (categorical and numerical) educational dataset.

III. Research Objectives

The objectives of this study are to:

1) Perform a thorough exploratory data analysis of the Students Performance dataset to understand score distributions, categorical group differences, and inter-subject correlations.
2) Engineer a meaningful multi-class target variable (writing performance category) from the continuous writing score while avoiding target leakage.
3) Design and implement a robust, leakage-free preprocessing pipeline for mixed categorical and numerical features.

4) Train and evaluate five supervised classification algorithms: Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors, and Support Vector Machine.
5) Validate model consistency using stratified k-fold cross-validation and optimize the best-performing model using hyperparameter tuning.
6) Compare all trained models using standard classification metrics and select the best-performing model.
7) Interpret the selected model's coefficients to identify the most influential predictors of writing performance.
8) Demonstrate the practical use of the final model by predicting the performance category of an unseen student record.

IV. Research Questions

This study is guided by the following research questions:

RQ1. How are mathematics, reading, and writing scores distributed across the student population, and how strongly are they correlated with one another?
RQ2. Do demographic and socioeconomic factors – gender, race/ethnicity, parental level of education, lunch type, and test preparation course completion – show a measurable relationship with writing performance category?
RQ3. Which supervised classification algorithm achieves the best predictive performance for the writing performance classification task on this dataset?
RQ4. Does feature scaling and hyperparameter tuning improve the performance of the Logistic Regression model beyond its default configuration?
RQ5. Which input features contribute most strongly to the final model's predictions, and are these consistent with patterns observed during exploratory analysis?

V. Literature Review / Related Work

Educational Data Mining (EDM) and Learning Analytics have become established research areas that apply data mining and machine learning techniques to educational datasets in order to understand and improve learning processes. A substantial body of work has used student demographic, behavioral, and academic-history data to predict outcomes such as final grades, dropout risk, and pass/fail status.

Classical supervised learning algorithms – including Logistic Regression, Decision Trees, Random Forests, K-Nearest Neighbors, and Support Vector Machines – have each been widely applied to student performance prediction tasks in prior studies, typically because these datasets are of moderate size, contain a mix of categorical and numerical attributes, and require interpretable outputs for use by non-technical stakeholders such as teachers and administrators. Ensemble methods such as Random Forest are frequently reported to improve robustness over single Decision Trees by reducing variance, while linear models such as Logistic Regression are valued for their interpretability through examinable coefficients. Prior work in this space has also consistently highlighted socioeconomic proxies (such as lunch-program type, used as an indicator of economic status) and test-preparation participation as recurring correlates of academic performance, along with parental education level.

The Students Performance dataset used in this study is a widely used benchmark dataset in introductory and applied educational-data-mining projects, most commonly used to predict or classify performance in mathematics, reading, or writing based on the other available attributes.

VI. Research Gap

While many prior studies predict a single overall performance score or a pass/fail outcome, fewer studies explicitly frame writing performance as an independent multi-class target while rigorously excluding the target subject's own score from the input features to prevent target leakage. In addition, many applied educational-data-mining projects report only a single model's accuracy rather than a systematic, like-for-like comparison of multiple algorithms under an identical preprocessing pipeline, train/test split, and evaluation protocol. This study addresses that gap by (i) constructing a leakage-free multi-class target for writing performance, (ii) evaluating five different classification algorithms under an identical, pipeline-based experimental protocol, (iii) applying stratified cross-validation and hyperparameter tuning to the best-performing model, and (iv) providing model-level interpretability through Logistic Regression coefficient analysis, rather than treating the model as a black box.

VII. Dataset Description

The dataset used in this study is the *Students Performance* dataset, containing 1,000 student records and 8 original attributes, with no missing values and no duplicate rows. Table I summarizes the dataset attributes.

TABLE I
Dataset Attribute Summary

Attribute	Type	Description / Categories
gender	Categorical	female, male
race/ethnicity	Categorical	group A – group E
parental level of education	Categorical	some high school, high school, some college, associate's, bachelor's, master's
lunch	Categorical	standard, free/reduced
test preparation course	Categorical	none, completed
math score	Numerical	0–100
reading score	Numerical	17–100
writing score	Numerical	10–100

Descriptive statistics for the three numerical score variables are given in Table II. The mean scores are broadly similar across subjects (math: 66.09, reading: 69.17, writing: 68.05), with comparable standard deviations of approximately 15 points.

The categorical attributes were also examined for class balance. Gender is nearly balanced (female: 518, male: 482). Race/ethnicity is dominated by group C (319) and group D (262), with group A the smallest (89). Parental education ranges from "some high school" (179) to "master's degree"

TABLE II
DESCRIPTIVE STATISTICS OF NUMERICAL SCORE VARIABLES

Statistic	Math	Reading	Writing
Count	1000	1000	1000
Mean	66.09	69.17	68.05
Std. Dev.	15.16	14.60	15.20
Min	0	17	10
25%	57	59	57.75
50% (Median)	66	70	69
75%	77	79	79
Max	100	100	100

(59, the smallest category). Lunch type is imbalanced toward standard (645) over free/reduced (355), and test preparation course is imbalanced toward "none" (642) over "completed" (358).

VIII. METHODOLOGY

The overall methodology follows the standard supervised machine learning workflow: data understanding and cleaning, exploratory data analysis, target variable construction, feature selection, a leakage-free train/test split, pipeline-based preprocessing, training of multiple candidate classification algorithms, validation via cross-validation, hyperparameter tuning of the best candidate, final evaluation on the untouched test set, model comparison, interpretability analysis, and final deployment-style prediction on an unseen student record. All preprocessing steps that could leak information from the test set (encoding, scaling) were implemented inside a single `scikit-learn Pipeline` object per model, so that these steps are fit only on the training data.

IX. SYSTEM / ML WORKFLOW

Fig. **??** (described textually below, as the workflow was implemented directly in code rather than as a separate diagram) summarizes the end-to-end system pipeline used in this study:

1) **Data Ingestion:** Load the raw CSV dataset into a pandas DataFrame.
2) **Data Quality Checks:** Inspect shape, data types, missing values, and duplicate rows.
3) **Exploratory Data Analysis:** Visualize score distributions, categorical group differences, correlations, and outliers.
4) **Target Engineering:** Compute the overall score, and convert the writing score into three ordinal performance categories (Low, Good, Excellent).
5) **Feature Selection:** Select demographic, socioeconomic, and non-target academic score features as model inputs.
6) **Train/Test Split:** Perform an 80/20 stratified split.
7) **Preprocessing Pipeline:** One-hot encode categorical features; optionally standardize numerical features.
8) **Model Training:** Fit five candidate classifiers using identical training data and preprocessing.
9) **Validation:** Apply 5-fold stratified cross-validation to the leading model.
10) **Hyperparameter Tuning:** Apply GridSearchCV to optimize the regularization strength of Logistic Regression.
11) **Evaluation and Comparison:** Compare all models on the held-out test set using accuracy and macro F1-score.
12) **Interpretability:** Analyze model coefficients to identify influential features.
13) **Deployment-style Prediction:** Predict the performance category of a new, unseen student record.

X. DATA PREPROCESSING

The dataset required minimal cleaning: an inspection of column data types, missing values, and duplicate rows confirmed 1,000 complete records with no missing values and no duplicate rows, so no imputation or deduplication was necessary. The five categorical attributes (gender, race/ethnicity, parental level of education, lunch, test preparation course) were transformed using one-hot encoding (`OneHotEncoder` with `handle_unknown="ignore"`), while the two numerical input features (math score, reading score) were either passed through unchanged or standardized using `StandardScaler`, depending on the model variant (Section XVII). All encoding and scaling logic was placed inside a `ColumnTransformer` nested within a `scikit-learn Pipeline`, ensuring that transformation parameters were learned exclusively from the training partition.

XI. EXPLORATORY DATA ANALYSIS (EDA)

EDA was performed to understand score distributions, target class balance, categorical group effects, and inter-variable correlation before model development.

A. Score Distributions

The distributions of math, reading, and writing scores were visualized using histograms with kernel density estimates (Figs. 1–3). All three subjects show a broadly unimodal distribution: most students score between approximately 50 and 85, with the highest concentration in the 60–80 range and a longer tail toward lower scores. A composite overall score (the mean of the three subject scores) was also computed and visualized (Fig. 4); its distribution mirrors the individual subject distributions, with a mean of 67.77 and standard deviation of 14.26.

B. Categorical Feature Relationships

Percentage-normalized bar charts (Figs. 5–9) were used to examine how writing performance category varies across categorical groups. Female students show a substantially higher rate of Excellent writing performance (33.2%) than male students (13.1%), while male students show a much higher rate of Low performance (38.8% vs. 18.1% for females). Students whose parents hold a master's degree show the highest Excellent rate (39.0%) and lowest Low rate (13.6%) among parental-education groups, while students with a high-school-only parental education background show the lowest Excellent rate (11.2%) and highest Low rate (39.3%). Students

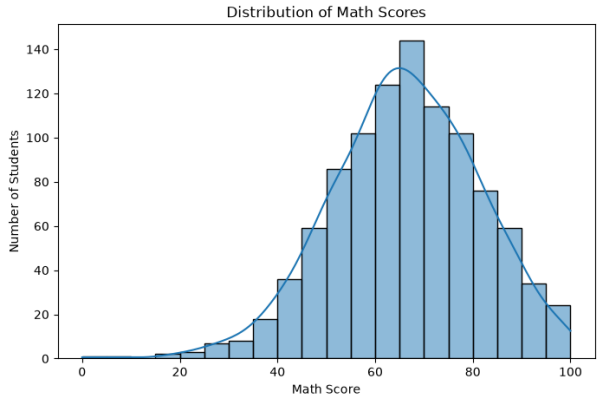


Fig. 1. Distribution of math scores.

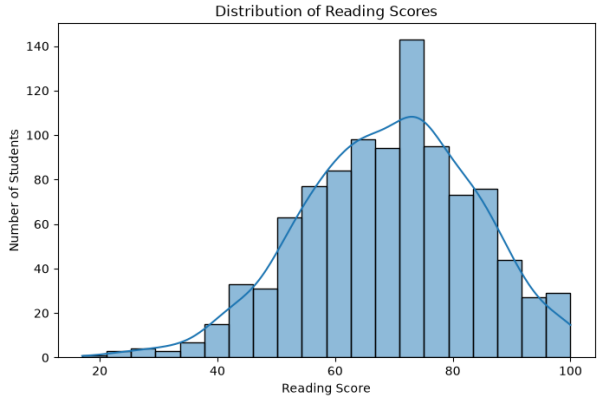


Fig. 2. Distribution of reading scores.

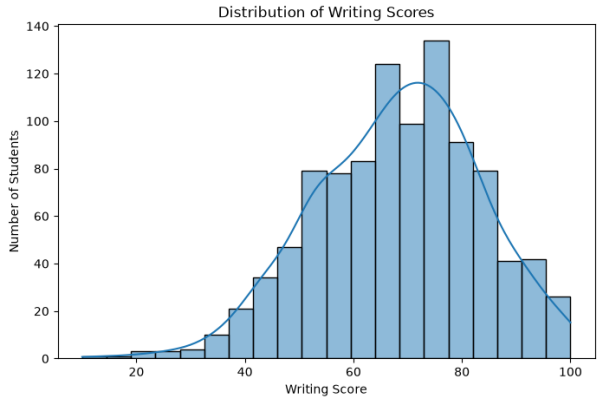


Fig. 3. Distribution of writing scores.

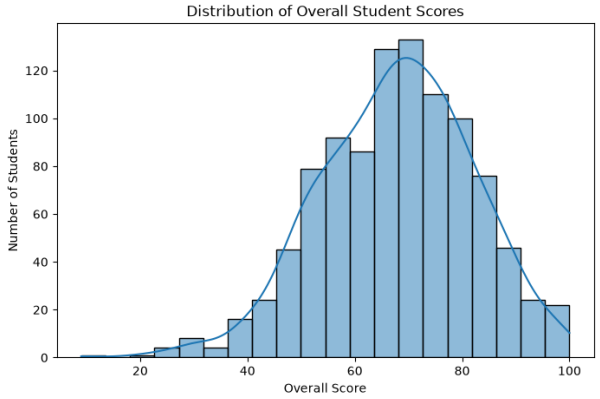


Fig. 4. Distribution of the overall (mean) student score.

with standard lunch show a higher Excellent rate (28.1% vs. 15.2%) and lower Low rate (21.2% vs. 40.6%) than students with free/reduced lunch. Students who completed the test preparation course show a markedly higher Excellent rate (36.9% vs. 16.0%) and lower Low rate (14.0% vs. 36.0%) than students who did not.

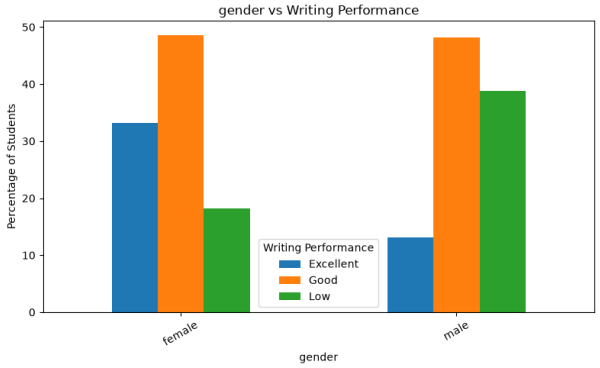


Fig. 5. Gender vs. writing performance category.

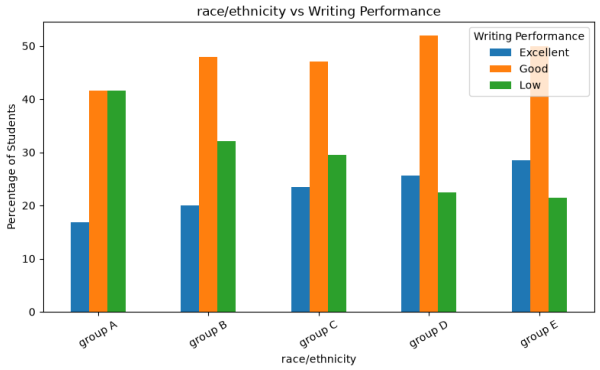


Fig. 6. Race/ethnicity vs. writing performance category.

C. Numerical Feature Relationships

Boxplots of math and reading scores across writing performance categories (Figs. 10–11) show a clear monotonic

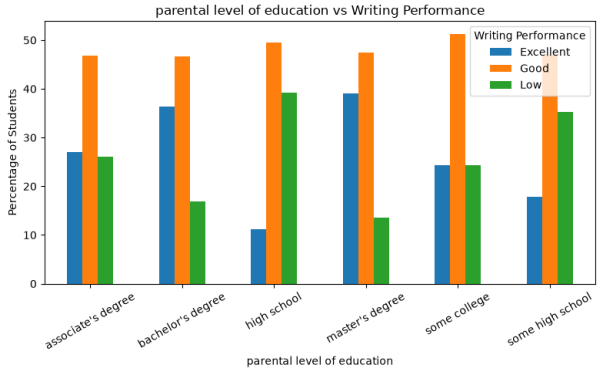


Fig. 7. Parental level of education vs. writing performance category.



Fig. 8. Lunch type vs. writing performance category.

relationship: median math and reading scores increase steadily from the Low group, through Good, to Excellent, confirming that performance in related subjects is strongly associated with writing performance.

D. Correlation Analysis

A Pearson correlation matrix (Fig. 12) was computed for the three subject scores. Reading and writing scores are very strongly correlated ($r = 0.955$), while math score correlates strongly with both reading ($r = 0.818$) and writing ($r = 0.803$). This strong inter-subject correlation motivated

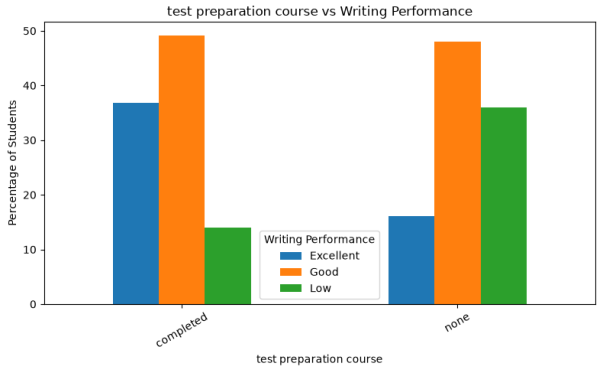


Fig. 9. Test preparation course completion vs. writing performance category.

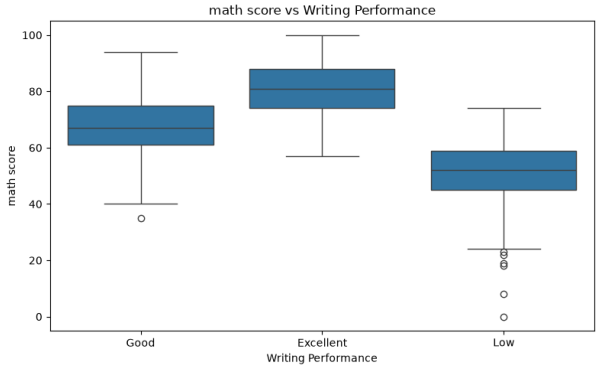


Fig. 10. Math score distribution across writing performance categories.

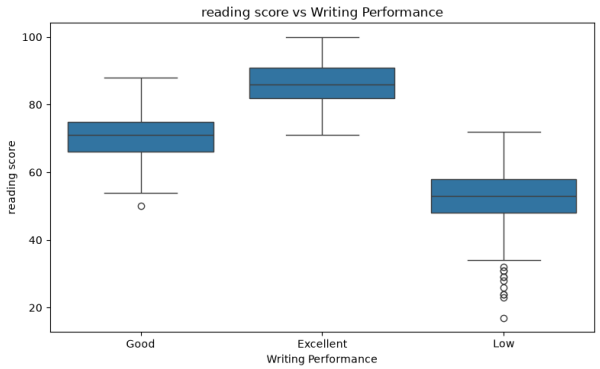


Fig. 11. Reading score distribution across writing performance categories.

the deliberate exclusion of the writing score itself from the input feature set, since including it (or a near-duplicate of it) would trivially leak the target.

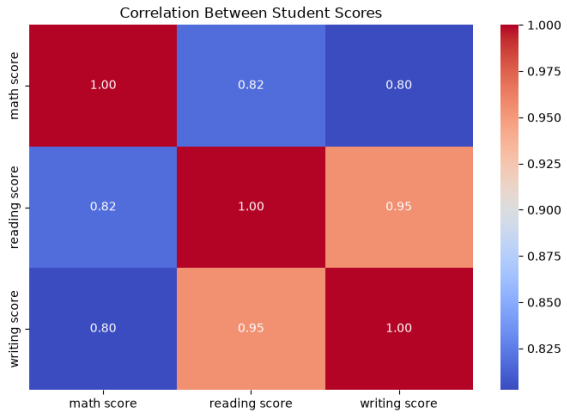


Fig. 12. Correlation heatmap of math, reading, and writing scores.

E. Outlier Analysis

Using the interquartile range (IQR) method, 8 potential outliers were identified in math scores (bounds: 27.00–107.00) and 6 in reading scores (bounds: 29.00–109.00). These were retained in the dataset, as unusually low scores may represent

genuine, valid student observations rather than data-entry errors.

XII. Feature Engineering / Feature Selection

Two engineered variables were constructed from the raw scores. First, an *overall score* was computed as the mean of the math, reading, and writing scores, used only for exploratory analysis of a possible three-tier performance scheme. Second, and central to this study, the *writing performance* target variable was engineered by discretizing the writing score into three ordinal categories: *Low* (writing score < 60), *Good* ($60 \leq$ writing score < 80), and *Excellent* (writing score ≥ 80). This produced a class distribution of Good: 484 (48.4%), Low: 281 (28.1%), and Excellent: 235 (23.5%) – a reasonably balanced multi-class target.

The final input feature set comprised seven features: gender, race/ethnicity, parental level of education, lunch, test preparation course, math score, and reading score. The writing score was deliberately excluded from the input features, since it was used to construct the target variable; including it would constitute target leakage and would produce an unrealistically inflated and unusable evaluation.

XIII. Model Development

Five supervised classification algorithms were developed, each wrapped in an identical `scikit-learn Pipeline` consisting of the shared preprocessing step followed by the classifier:

- **Logistic Regression** (`max_iter=1000`): a linear probabilistic classifier that estimates class membership probabilities via the softmax/logit function.
- **Decision Tree Classifier** (`random_state=42`): a rule-based, non-linear model that recursively partitions the feature space.
- **Random Forest Classifier** (`n_estimators=100`, `random_state=42`): an ensemble of 100 decision trees combined via majority voting to reduce variance.
- **K-Nearest Neighbors** (`n_neighbors=5`): an instance-based classifier that assigns the majority class among the 5 nearest training observations.
- **Support Vector Machine** (default RBF kernel `SVC()`): a margin-maximizing classifier that separates classes using a non-linear kernel-induced boundary.

All models were trained on the identical 800-record training partition, using the identical set of seven input features and identical preprocessing (one-hot encoding of categorical features), ensuring a fair, like-for-like comparison.

XIV. Experimental Setup

All experiments were implemented in Python using `pandas`, `scikit-learn`, `matplotlib`, and `seaborn`. The dataset (1,000 records) was split into training (800 records, 80%) and test (200 records, 20%) partitions using `train_test_split` with `stratify=y` and `random_state=42`, ensuring that the class proportions of Low, Good, and Excellent were preserved in both partitions (training: 48.4%/28.1%/23.5%; test: 48.5%/28.0%/23.5% for Good/Low/Excellent respectively). This fixed random seed ensures the experiment is reproducible. All five models were trained and evaluated under this identical split and identical preprocessing pipeline.

XV. Model Validation

To assess the stability of the leading model (Logistic Regression) beyond a single train/test split, 5-fold stratified cross-validation (`StratifiedKFold`, `shuffle=True`, `random_state=42`) was performed on the training partition only, with the test set held out entirely. The default (unscaled) Logistic Regression pipeline achieved cross-validation fold accuracies of [0.869, 0.850, 0.863, 0.831, 0.881], with a mean accuracy of 0.8587 and a standard deviation of 0.0170, indicating stable performance across folds. This training also triggered convergence warnings from the `lbfgs` solver, motivating the standardized variant described below.

XVI. Hyperparameter Tuning

To address the solver convergence warnings and explore whether performance could be improved, numerical features were standardized using `StandardScaler` (categorical features remained one-hot encoded), and `max_iter` was increased to 2000. The resulting scaled Logistic Regression model achieved a 5-fold cross-validation mean accuracy of 0.8663 (std. 0.0196), fold scores: [0.869, 0.863, 0.881, 0.831, 0.888]. `GridSearchCV` was then used to tune the inverse regularization strength `C` over the grid $\{0.01, 0.1, 1, 10, 100\}$ using the same 5-fold stratified cross-validation and accuracy scoring. The best parameter identified was $C = 1$ (the library default), with a best cross-validation accuracy of 0.8663, indicating that the default regularization strength was already optimal for this dataset and that further tuning of C did not yield an improvement.

XVII. Model Evaluation

Each of the five trained models was evaluated on the untouched 200-record test set using accuracy, per-class precision/recall/F1-score, and macro-averaged F1-score. Table III shows the detailed classification report for the best-performing model (default Logistic Regression), and Figs. 13 and 14 show its confusion matrix before and after final selection (identical model, shown for both initial and final evaluation steps in the original workflow).

TABLE III
Classification Report – Logistic Regression (Test Set)

Class	Precision	Recall	F1	Support
Excellent	0.82	0.87	0.85	47
Good	0.89	0.88	0.89	97
Low	0.95	0.93	0.94	56
Accuracy		0.89		200
Macro avg	0.89	0.89	0.89	200
Weighted avg	0.89	0.89	0.89	200

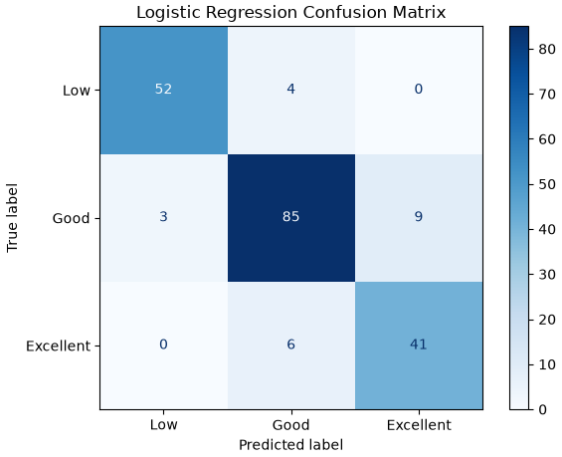


Fig. 13. Confusion matrix – Logistic Regression on the test set.

The scaled and hyperparameter-tuned Logistic Regression model achieved a slightly lower test accuracy of 88.0% (macro F1: 0.8790) than the original unscaled default model's 89.0% (macro F1: 0.8892), indicating that, for this particular dataset and split, scaling and regularization tuning did not translate into an improvement on the held-out test set, even though it resolved the solver's convergence warning and offered marginally more stable cross-validation behavior.

XVIII. Model Comparison

Table IV compares all five classification algorithms on the held-out test set. Logistic Regression achieved the highest test accuracy (0.8900) and macro F1-score (0.8892), followed closely by Random Forest (0.8800 / 0.8782), K-Nearest Neighbors (0.8700 / 0.8703), Support Vector Machine (0.8600 / 0.8613), and Decision Tree, which had the lowest performance (0.8300 / 0.8306).

TABLE IV
Model Comparison on the Held-out Test Set

Model	Accuracy	Macro F1
Logistic Regression	**0.8900**	**0.8892**
Random Forest	0.8800	0.8782
K-Nearest Neighbors	0.8700	0.8703
Support Vector Machine	0.8600	0.8613
Decision Tree	0.8300	0.8306

The consistent ranking of Logistic Regression at the top of both metrics, combined with its cross-validated stability (Section XV), supports its selection as the final model. The relatively strong performance of a simple linear model over more flexible non-linear models (Decision Tree, Random Forest, KNN, SVM) suggests that the relationship between the selected input features and writing performance is largely linear and additive in nature, at least after one-hot encoding of categorical variables.

XIX. Final Model Selection

Based on the model comparison in Section XVIII, the original (unscaled, default-hyperparameter) **Logistic Regression** pipeline was selected as the final model, achieving a test accuracy of 89.0% and a macro F1-score of 0.8892 – the best result among all five algorithms and configurations evaluated, including the scaled and hyperparameter-tuned variant. Its final confusion matrix on the test set is shown in Fig. 14: of 56 Low-category students, 52 were correctly classified; of 97 Good-category students, 85 were correctly classified; and of 47 Excellent-category students, 41 were correctly classified. Most misclassifications occurred between the adjacent Good and Excellent, or Good and Low categories, consistent with the ordinal nature of the target variable, rather than between the non-adjacent Low and Excellent categories.

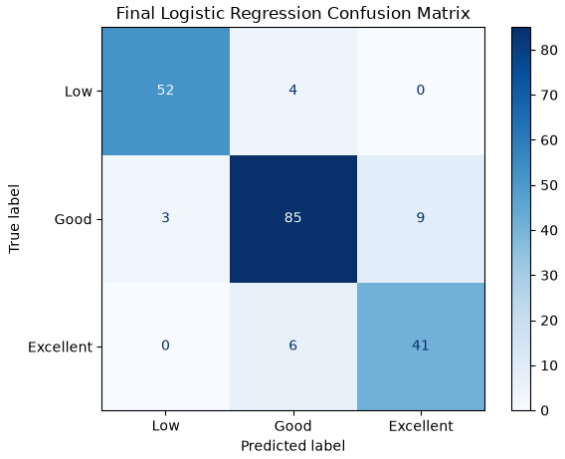


Fig. 14. Final confusion matrix of the selected Logistic Regression model.

To interpret the final model, the Logistic Regression coefficients were extracted per class after one-hot encoding (19 transformed features across 3 classes). Table V lists the top overall predictors, ranked by the mean absolute coefficient across the three classes. Gender (female / male) was the single most influential predictor, followed by parental education (bachelor's degree), and test preparation course completion (completed / none) – consistent with the categorical group differences observed during EDA (Section XI-B).

TABLE V
Top 8 Most Influential Features (Mean Absolute Coefficient)

Feature	Mean |Coefficient|
gender: female	0.9734
gender: male	0.8646
parental education: bachelor's degree	0.6776
test preparation course: completed	0.6587
test preparation course: none	0.5499
race/ethnicity: group D	0.4107
parental education: some high school	0.3760
race/ethnicity: group E	0.3731

XX. Sample / Unseen Data Prediction

To demonstrate the practical, deployment-style use of the final model, a new, previously unseen student record was constructed: a female student in race/ethnicity group C, whose parent completed some college, with standard lunch, who completed the test preparation course, with a math score of 75 and a reading score of 80 (writing score intentionally omitted, since it is the value being predicted). The final Logistic Regression pipeline predicted a writing performance category of **Excellent** for this student, with class probabilities of 82.53% (Excellent), 17.47% (Good), and 0.00% (Low). This illustrates how the trained pipeline can be applied directly to new, real-world student records using only readily available demographic and academic information.

XXI. Results

Summarizing the experimental results: (1) all five classifiers exceeded 83% test accuracy on a well-balanced three-class target, indicating that the selected features carry substantial predictive signal for writing performance; (2) Logistic Regression was the best-performing model overall (89.0% accuracy, 0.8892 macro F1), ahead of Random Forest, KNN, SVM, and Decision Tree, in that order; (3) 5-fold cross-validation confirmed the stability of the Logistic Regression model (mean accuracy 0.8587–0.8663 depending on preprocessing variant, low standard deviation); (4) feature scaling and `GridSearchCV`-based hyperparameter tuning of C did not improve test-set performance beyond the untuned default model; (5) gender, parental education, and test preparation course completion emerged as the most influential predictors; and (6) the final model produced sensible, well-calibrated probability estimates on an unseen student record.

XXII. Discussion

The strong performance of Logistic Regression relative to the more flexible non-linear models is notable. With only seven input features (five categorical, two numerical) and a modest dataset size of 1,000 records, the relationship between inputs and the writing-performance target appears close to linearly separable after one-hot encoding, which limits the advantage that non-linear models such as Random Forest, KNN, and SVM might otherwise offer, and increases the risk of those models overfitting on a comparatively small, largely categorical feature space. The strong correlation between reading and writing scores ($r = 0.955$) explains why reading score retained meaningful predictive weight even though writing score itself was excluded. The pronounced gender gap and test-preparation-course effect observed in the raw EDA were directly reflected in the model's largest coefficients, giving confidence that the model has learned genuine, interpretable patterns in the data rather than spurious correlations. The fact that hyperparameter tuning left the default regularization strength ($C = 1$) unchanged as optimal further suggests that the default Logistic Regression configuration was already well suited to this problem, and that additional model complexity was not required.

XXIII. Limitations

Several limitations should be acknowledged. First, the dataset is comparatively small (1,000 records) and originates from a single, unspecified educational context, which may limit the generalizability of the findings to other student populations. Second, the categorical group definitions (e.g., race/ethnicity groups A–E) are anonymized in the source dataset, which limits the sociological interpretability of some coefficients. Third, only a single train/test split (with cross-validation applied to one model) was used for the primary comparison across all five algorithms; a fully nested cross-validation across all algorithms would provide a more statistically robust comparison. Fourth, hyperparameter tuning was restricted to a single parameter (C) of a single model (Logistic Regression); the other four algorithms were evaluated only with their default hyperparameters, which may understate their true achievable performance. Finally, the three-category discretization of writing score (Low / Good / Excellent) introduces fixed threshold boundaries (60 and 80) that, while reasonable, are somewhat arbitrary and could affect class boundary predictions for borderline students.

XXIV. Conclusion

This paper presented a complete, end-to-end machine learning pipeline for classifying student writing performance into Low, Good, and Excellent categories using demographic, socioeconomic, and non-target academic features. After thorough exploratory data analysis, careful leakage-free feature and target engineering, and a systematic comparison of five supervised classification algorithms under an identical experimental protocol, Logistic Regression was identified as the best-performing and most stable model, achieving 89.0% test accuracy and a macro F1-score of 0.8892. Coefficient-based interpretability analysis confirmed that gender, parental education, and test preparation course completion are the strongest predictors of writing performance, consistent with patterns already visible in the exploratory analysis. The final model was successfully applied to an unseen student record, correctly demonstrating its practical, deployment-ready usability. These findings suggest that simple, interpretable linear classifiers remain highly competitive for small-to-moderate, largely categorical educational datasets of this type.

XXV. Future Scope

Future work could extend this study in several directions: (1) applying nested cross-validation and hyperparameter tuning to all five algorithms, rather than Logistic Regression alone, for a more rigorous comparison; (2) exploring additional ensemble and boosting methods such as Gradient Boosting, XGBoost, or LightGBM; (3) extending the classification target to mathematics and reading performance, or to a joint multi-output prediction of all three subjects; (4) validating the approach on larger, multi-institution datasets to test generalizability; (5) incorporating additional features such as attendance, study time, or prior-year grades where available; and (6) deploying

the final model as an interactive early-warning dashboard for use by teachers and academic advisors.

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APPENDIX A
ADDITIONAL EXPERIMENTAL DETAILS

This appendix provides supporting details from the final machine learning experiment that are not repeated in the main discussion.

A. Target Class Distribution

The writing-performance target consisted of three categories: Low, Good, and Excellent. The complete dataset contained 484 Good students (48.4%), 281 Low students (28.1%), and 235 Excellent students (23.5%).

B. Final Model Configuration

The final selected model was Logistic Regression with a maximum of 1000 iterations. The dataset was divided into 800 training records and 200 testing records using an 80:20 stratified split with `random_state=42`. Categorical variables were transformed using one-hot encoding, while math and reading scores were retained as numerical inputs.

Five-fold stratified cross-validation was used to assess model stability. The final model achieved 89.0% test accuracy and a macro F1-score of 0.8892.

C. Unseen Student Prediction

For the final demonstration, an unseen student record was supplied to the selected model using the following attributes:

D. Unseen Student Prediction

For the final demonstration, an unseen student record was supplied to the selected model using the following attributes:

Feature	Value
Gender	Female
Race/ethnicity	Group C
Parental education	Some college
Lunch	Standard
Test preparation	Completed
Math score	75
Reading score	80

The model predicted the writing-performance category as **Excellent**. The predicted class probabilities were 82.53% for Excellent, 17.47% for Good, and 0.00% for Low.

The model predicted the writing-performance category as **Excellent**. The predicted class probabilities were 82.53% for Excellent, 17.47% for Good, and 0.00% for Low.